Coevolutionary balance of resting-state brain networks in autism

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Short title: Coevolutionary balance in adult male autism

Abstract

Signed functional connectivity and regional low-frequency BOLD fluctuations offer complementary descriptions of resting-state organization. We examined their joint alignment using a static Hamiltonian inspired by coevolutionary balance in a male-only adult sample from ABIDE I. Downloaded preprocessed regional time series and fALFF maps were analyzed with and without global signal regression (GSR). Age and IQ eligibility yielded 140 participants (69 autistic participants and 71 controls). Measurement coverage defined a fixed common set of 106 regions; site-adjusted group models included 136 participants. In a separate male-control baseline, mean empirical Hamiltonian values were lower than their null expectations under unrestricted and network/hemisphere-blocked state permutations. Hamiltonian values did not show a detectable diagnosis effect after adjustment for site, age, full-scale IQ, and mean framewise displacement (GSR: coefficient 1.43, 95% CI −17.17 to 20.04; no-GSR: 2.06, 95% CI −12.39 to 16.50; both Holm-adjusted $p = 1.00$). ComBat, pooled energy-trimming, and larger-node-set sensitivities did not identify a corrected diagnosis effect. Equivalence tests were inconclusive. Exploratory site-held-out classification yielded balanced accuracies of 0.457–0.558, with all conditional confidence intervals including 0.5. A broader exploratory selection-method search did not establish a reliable performance gain. Clinical analyses and motion- and mean-connectivity-adjusted efficiency analyses provided no corrected evidence of an association. These results do not establish group equivalence, diagnostic utility, or an autism-specific energy–efficiency relationship. Interpretation is restricted to the selected adult males, the coverage-limited network, and the documented limits of the downloaded derivatives.

Introduction

Resting-state functional MRI provides several descriptions of brain organization. Functional connectivity (FC) measures statistical dependence between regional BOLD time series, whereas fractional amplitude of low-frequency fluctuations (fALFF) describes the relative contribution of low-frequency fluctuations to the sampled spectrum [1]. Neither quantity directly measures neuronal firing rate. Regional fALFF differences have been reported in adult autism [2], but such regional effects need not produce a difference in a scalar summary combining fALFF ranks and FC.

Signed networks retain positive and negative associations, whose treatment can materially affect network summaries [3]. Structural balance theory describes consistency among signed relationships [4, 5]. Coevolutionary models introduce nodal states as well as signed interactions [6, 7]. Here, these ideas motivate a static alignment statistic. We do not estimate a dynamical coevolutionary process, metabolic energy, or excitation–inhibition balance from BOLD data.

Our earlier ABIDE structural-balance analysis studied triadic relationships in children and adolescents aged 6–18 without GSR [8]. Its statistic and developmental population differ from the present adult nodal-state Hamiltonian. Likewise, energy-landscape analyses of autism model time-varying state occupancy and transitions [9]; a static alignment statistic cannot reproduce or refute a finding about those transitions.

Multisite FC studies show substantial heterogeneity and limited generalization across acquisition sites [10]. Site adjustment and harmonization therefore require explicit treatment, particularly when diagnosis and site are unbalanced. Protecting diagnosis in a harmonization model does not guarantee unbiased downstream inference [11]. We use raw adjusted features for primary inference and treat ComBat as a sensitivity analysis.

We ask whether a coverage-defined alignment statistic differs by diagnosis in adult males with observed full-scale IQ (FIQ) above 80. We compare GSR and no-GSR derivatives, quantify uncertainty, examine the role of mean FC and motion in efficiency associations, and assess exploratory site-held-out prediction. The node and participant definitions were fixed using demographic criteria and measurement QC before corrected group-effect testing; they are revision-stage choices, not an original preregistration.

The joint statistic addresses a question that neither regional amplitude nor FC alone answers: whether the ordering of regional fluctuations agrees with the sign and strength of pairwise coupling. A negative connection between opposite relative states contributes in the same favorable direction as a positive connection between matching states. This symmetry is central to the mathematical construction; it should

not be confused with a physiological claim that a negative BOLD correlation implements inhibition. Retaining signed magnitudes also means that a few strong connections can outweigh many weak connections. Thus, configuration counts and their weighted Hamiltonian provide related but distinct descriptions of network organization.

Autism-related connectivity findings are not uniformly absent or uniformly reproducible across datasets. Large-cohort work has identified reproducible connectivity alterations while also documenting substantial overlap between individuals with and without autism [12]. Other multisite analyses emphasize variation across sites and analytic choices [10]. These findings motivate an explicit separation of group inference from prediction: a population association is not automatically an individual diagnostic marker, and unsuccessful classification does not disprove all regional abnormalities. A global alignment summary may be insensitive to heterogeneous regional effects that cancel when aggregated.

We retain the original hierarchy of questions: alignment against randomized states, whole-network configuration and energy comparisons, intra- and inter-network comparisons, relationships with complementary network measures, clinical correlations and partial least squares, and classification. GSR/no-GSR comparisons are carried through each level. The revised analysis changes the measurement and validation definitions where the original records required clarification, while retaining these scientific questions rather than replacing them with a different study.

Results

Participant and measurement scope

Figure 1 separates demographic eligibility, feature availability, and site-adjusted inference. Table 1 distinguishes the 140 eligible participants from the 136 modeled participants. The nine-site model includes an ASD-only site, CALTECH ($n = 9$); its diagnosis effect is identified through the remaining mixed-diagnosis sites, under the common-effect regression assumption. Broader-181 and full-200 designs have 101 (47/54) and 64 (30/34) adjusted-model participants, respectively.

Fig 1. Participant flow and derivative-based analysis. The requested pool contains 182 adult-source participants. Age and observed IQ eligibility yields 140 male participants; all share 106 coverage-qualified nodes. Raw adjusted group models include 136 participants after excluding sites with fewer than three retained participants. Classification and individual null analyses retain all 140. MRI preprocessing and voxelwise fALFF computation occurred upstream.

Table 1. Participant characteristics in the two primary analysis samples.

Sample	Group	N	Age, years	FIQ	Mean FD, mm
Eligible	ASD	69	22.23 (3.42)	111.23 (12.85)	0.083 (0.045)
Eligible	Control	71	23.00 (3.35)	115.63 (10.98)	0.074 (0.032)
Adjusted model	ASD	67	22.16 (3.42)	111.22 (12.77)	0.081 (0.044)
Adjusted model	Control	69	22.86 (3.30)	115.66 (11.14)	0.073 (0.032)

Continuous values are mean (SD). All participants were recorded male. Eligibility is age 18–30 inclusive and observed FIQ > 80. No handedness restriction or missing-IQ imputation is applied. The adjusted-model subset excludes four participants at sites with fewer than three eligible participants.

Null-model comparisons of coevolutionary alignment

Male-control baseline. We first examined the 71 eligible male controls as a separate non-autistic baseline, rather than pooling them with ASD. Each control contributes 1,000 state permutations per pipeline and null scheme. Mean empirical H is -91.39 with GSR and -66.62 without GSR. The corresponding unrestricted null means are 1.49 and 17.11; mean empirical-minus-null differences are -92.88 and -83.73, with descriptive site-bootstrap 95% intervals of -103.90 to -79.89 and -89.79 to -77.66, respectively.

With network/hemisphere-constrained states, the null means are -41.35 and -18.85, and the empirical-minus-null differences are -50.04 (95% interval -54.14 to -44.03) and -47.76 (-53.11 to -42.16). Table 2 and Fig. 2 show these comparisons; individual null z-scores are plotted in S1 Appendix. All 71 controls in each pipeline have empirical H below their unrestricted null mean. Under the blocked null, this holds for 69/71 with GSR and 67/71 without GSR. After exploratory BH correction over all 560 individual participant/pipeline/null comparisons, the numbers with lower-tail $q < .05$ are 61/71 and 64/71 for the unrestricted null and 36/71 and 50/71 for the blocked null (GSR/no-GSR).

Fig 2. Empirical alignment relative to individual null means. Each point is empirical H minus the participant-specific mean of 1,000 permutations. Panels separate pipelines and unrestricted versus network/hemisphere-blocked reassignment. ASD is magenta and controls dark green. FC is fixed; these are state permutations, not topology-rewired graphs. Negative values indicate lower empirical H than the specified null expectation.

Thus empirical control state assignments tend to align with their own FC more strongly than the randomized assignments under both specified nulls, with substantial dependence on the constraints. This is evidence about the chosen alignment statistic and randomizations, not proof of dynamical equilibrium, a comparison with topology-rewired graphs, or an autism-specific effect.

Table 2. Empirical alignment against state-permutation null configurations.

Group	Pipeline	Null	N	Mean H	Mean null H	Mean z	q<.05 count
ASD	GSR	B	69	-93.173	-43.546	-2.331	40
ASD	GSR	U	69	-93.173	1.611	-4.705	61
ASD	noGSR	B	69	-64.250	-20.849	-2.591	46
ASD	noGSR	U	69	-64.250	16.297	-4.917	60
Control	GSR	B	71	-91.392	-41.354	-2.313	36
Control	GSR	U	71	-91.392	1.486	-4.690	61
Control	noGSR	B	71	-66.618	-18.854	-2.760	50
Control	noGSR	U	71	-66.618	17.109	-5.159	64

U: unrestricted state permutations; B: network/hemisphere-blocked permutations. Each person has 1,000 permutations under each scheme. BH correction here spans all 560 person/pipeline/null comparisons; the counts do not test an ASD–control difference.

ASD and across-diagnosis summaries. All 140 participants per pipeline have finite null scores under both randomizations. Within ASD alone ($n = 69$), mean unrestricted z is $-4.705/-4.917$ (GSR/no-GSR), and blocked mean z is $-2.331/-2.591$. With exploratory BH across all 560 participant/pipeline/null comparisons, the ASD lower-tail $q < .05$ counts are 61/69 and 60/69 for unrestricted states, and 40/69 and 46/69 for blocked states. These complements retain the original ASD as well as control null question and do not test diagnosis differences. The corresponding across-diagnosis summaries are retained for completeness. Mean unrestricted z is $-4.70/-5.04$ (GSR/no-GSR), compared with $-2.32/-2.68$ under network/hemisphere blocks. These values show dependence on the randomization scheme, not diagnosis discrimination or preservation of full spatial structure. Individual scores and probabilities are in S2 Data.

Whole-network coevolutionary energy and motif composition

Primary adjusted coefficients are 1.432 with GSR (95% CI -17.173 to 20.038, $p = .879$) and 2.056 without GSR (95% CI -12.387 to 16.499, $p = .779$); both Holm-adjusted $p = 1.000$ (Table 3, Fig. 3). No valid secondary group test survives its specified correction in any raw or ComBat design. The trimming sensitivities likewise do not identify a corrected H difference. Larger-node results are in S1 Appendix.

Fig 3. Global feature distributions in both processing pipelines. Points and boxes describe the untrimmed 140-person feature cohort: 69 ASD and 71 controls. ASD is magenta and controls dark green. Seven available global quantities include H, four pairwise configuration proportions and two absolute-weight efficiencies. Adjusted inference uses 136 participants; distribution overlap is not an equivalence test.

Table 3. Adjusted diagnosis coefficients for primary-network H.

Pipeline	Model	N	Coefficient	95% CI	p	Holm p
GSR	Raw adjusted	136	1.432	-17.17, 20.04	0.879	1.000
GSR	ComBat	136	1.139	-17.15, 19.42	0.902	1.000
GSR	Pooled trim	135	-2.613	-19.56, 14.34	0.761	1.000
noGSR	Raw adjusted	136	2.056	-12.39, 16.50	0.779	1.000
noGSR	ComBat	136	2.247	-11.81, 16.30	0.752	1.000
noGSR	Pooled trim	135	3.500	-10.77, 17.77	0.628	1.000

Coefficients compare ASD with controls, adjusting for site, age, FIQ and mean FD. HC3 intervals use a t reference. Holm correction covers the two pipelines within each model version. ComBat and pooled trimming are exploratory sensitivities.

Figure 3 displays all seven valid global quantities for the untrimmed 140-person feature cohort. Within- and between-network effect heatmaps below show all prespecified cortical blocks rather than selecting nominally significant examples.

All six ComBat design/pipeline pairs complete with finite, changed feature values. Primary descriptive site partial R^2 decreases from .056 to .004 with GSR and from .047 to .005 without GSR. Maximum primary leverage is .352; maximum Cook's distance is .080 or smaller across these four raw/ComBat fits. These checks do not establish perfect harmonization or independence of participants within sites. Excluding CALTECH yields 127 participants (58 ASD, 69 controls) at eight sites; raw adjusted coefficients remain imprecise (GSR: 1.115, 95% CI −17.531 to 19.762; no-GSR: 1.660, 95% CI −12.775 to 16.096; both Holm $p = 1.000$).

Primary unadjusted Hedges' g is −.024 (bootstrap 95% CI −.384 to .315) with GSR and .072 (−.268 to .412) without GSR. Approximate two-sample 80%-power minimum detectable d is .484 at $\alpha = .05$, or .534 at $\alpha = .025$. Equivalence tests are inconclusive ($p = .193$ and .223). Nonsignificant contrasts therefore do not demonstrate absence of a meaningful diagnosis effect.

Network-specific coevolutionary metrics

Table 4 reports block-energy contrasts. We retain the original within-network question: whether ASD and control participants differ in mean FC (AvgWeight), mean state-weighted FC (AvgFalffW), or block Hamiltonian contribution (EnergyBlock) within each of the seven Yeo systems. The primary graph permits 18 tests per pipeline; three within-limbic quantities are unavailable because only one limbic node remains. All available blocks are shown in Fig. 4, with complete estimates in S2. No comparison survives BH across the 36 pipeline/within-network tests. In GSR, the smallest raw probability is .073. Without GSR, somatomotor mean FC is nominally lower in ASD (coefficient −.0683, $p = .0371$), as is default-network mean FC (−.0519, $p = .0432$); both become $q = .392$ and are not corrected findings.

Table 4. Within-network energy diagnosis coefficients.

Network	GSR beta	GSR 95% CI	GSR q	No-GSR beta	No-GSR 95% CI	No-GSR q
VIS	3.451	-0.51, 7.41	0.392	6.746	-0.27, 13.76	0.392
SM	-2.877	-6.74, 0.99	0.469	-2.712	-8.88, 3.46	0.830
DA	0.584	-0.83, 2.00	0.830	0.782	-1.47, 3.03	0.875
SV	-0.301	-2.37, 1.77	0.875	-0.496	-3.38, 2.39	0.875
LIM	–	–	–	–	–	–
FP	-0.620	-2.64, 1.40	0.875	-0.399	-3.88, 3.08	0.875
DMN	0.182	-1.46, 1.82	0.875	0.473	-2.16, 3.10	0.875

Raw adjusted models use 136 participants. BH spans both pipelines and all available within-network measures (36 tests). Limbic energy is unavailable because only one node is assigned. VIS: visual; SM: somatomotor; DA: dorsal attention; SV: salience/ventral attention; LIM: limbic; FP: frontoparietal; DMN: default mode. Other block measures are shown in Fig. 4 and S2 Data.

Fig 4. Within-network diagnosis comparisons. Cells show the raw adjusted diagnosis coefficient divided by its HC3 standard error for mean FC, mean state-weighted FC and block energy. Both pipelines use 136 participants. Gray within-limbic cells are unavailable because only one limbic node remains. None of the 36 valid tests survives BH correction. Color encodes signed t statistics, not significance.

Inter-network connectivity and energy

Table 5 reports all mean-FC contrasts. We also retain all 21 unordered network pairs and all three original measures, yielding 63 tests per pipeline. Fig. 5 shows every block, including nonsignificant ones. No comparison survives BH across the 126 pipeline/inter-network tests. GSR salience/ventral-attention–limbic state-weighted FC and its corresponding block energy are nominally different ($p = .0315$, $q = .756$). They are proportional descriptions of the same block, not two independent effects. No no-GSR inter-network test reaches $p < .05$; its smallest raw probability is .0508. These results address spatially specific diagnosis effects separately from the global Hamiltonian and the within-group associations.

Table 5. Between-network mean-FC diagnosis coefficients.

Pair	GSR beta	GSR p	GSR q	No-GSR beta	No-GSR p	No-GSR q
VIS–SM	-0.0059	0.738	0.893	-0.0523	0.082	0.756
VIS–DA	-0.0060	0.715	0.893	-0.0573	0.051	0.756
VIS–SV	0.0093	0.466	0.874	-0.0459	0.098	0.756
VIS–LIM	0.0289	0.228	0.835	-0.0388	0.318	0.835
VIS–FP	0.0117	0.288	0.835	-0.0381	0.143	0.756
VIS–DMN	-0.0052	0.684	0.877	-0.0531	0.053	0.756
SM–DA	-0.0115	0.506	0.877	-0.0486	0.139	0.756
SM–SV	0.0040	0.781	0.922	-0.0324	0.257	0.835
SM–LIM	-0.0110	0.666	0.877	-0.0592	0.098	0.756
SM–FP	0.0104	0.472	0.874	-0.0233	0.443	0.874
SM–DMN	0.0195	0.143	0.756	-0.0243	0.386	0.859
DA–SV	-0.0006	0.968	0.987	-0.0424	0.168	0.756
DA–LIM	-0.0284	0.307	0.835	-0.0678	0.097	0.756
DA–FP	-0.0010	0.940	0.987	-0.0365	0.186	0.780
DA–DMN	-0.0027	0.876	0.968	-0.0443	0.155	0.756
SV–LIM	-0.0132	0.587	0.877	-0.0603	0.078	0.756
SV–FP	-0.0123	0.244	0.835	-0.0450	0.115	0.756
SV–DMN	0.0139	0.258	0.835	-0.0322	0.273	0.835
LIM–FP	-0.0323	0.186	0.780	-0.0548	0.136	0.756
LIM–DMN	0.0251	0.335	0.835	-0.0146	0.689	0.877
FP–DMN	0.0039	0.783	0.922	-0.0294	0.289	0.835

Raw adjusted models use 136 participants. BH spans all three between-network measures in both pipelines (126 tests); q values therefore refer to the full family, not just this mean-FC table. Network abbreviations follow Table 4. State-weighted FC and energy are included in Fig. 5 and S2 Data.

Fig 5. Between-network diagnosis comparisons. All 21 cortical network pairs and all three block measures are shown in both pipelines. Cells show HC3 t statistics from raw adjusted models. No test survives BH across the 126 valid comparisons. Mean state-weighted FC and block energy are proportional for a fixed block and do not provide independent effects.

Relationships between H and other network metrics

To preserve the original descriptive analysis, we correlate H with all 87 other available nonlegacy features within ASD and controls separately, in both pipelines. All eligible participants are included (69 ASD/71 controls). Under the original per-group Pearson BH convention, 26 GSR-ASD, 21 GSR-control, and 12 no-GSR-ASD associations survive $q < .05$; none survives in no-GSR controls (Table 6). Figure 6 restores the original top-20 presentation, while S2 contains every test, not just ranked results.

Fig 6. Descriptive relationships between H and other network metrics. The top 20 absolute Pearson correlations among all 87 other available features are shown for each diagnosis and pipeline. Stars indicate BH q below .05 within the 87-test panel. Red and blue bars denote positive and negative correlations; panel headings retain diagnosis colors. W denotes mean FC, F state-weighted FC, and E block energy. Shared terms and proportional F/E pairs create mathematical dependence. These are within-group associations, not diagnosis contrasts.

Examples include GSR-ASD H versus somatomotor–default block energy ($r = .434$, $q = .00472$) and no-GSR-ASD H versus frontoparietal–default block energy ($r = .473$, $q = .000890$). EnergyBlock and AvgFalffW are proportional within a fixed block, and both share the state-weighted terms entering H;

these associations are therefore not independent validation. Among mean-FC features, GSR controls show an association with somatomotor–default AvgWeight ($r = -.315$, $q = .0346$), still derived from the same FC matrix. Significance in one group alone does not establish a group difference in correlation.

The descriptive raw H–global-efficiency correlations are .022 in GSR-ASD and .109 in no-GSR-ASD, neither significant; corresponding control correlations are .100 and .140. The adjusted analysis below addresses the additional question of association conditional on mean FC, motion, age, IQ, and site. Matched-participant raw/ComBat comparisons and alternative pooled corrections are retained in S2, without selecting the version with the most significant associations.

Table 6. Within-group associations between H and the other network metrics.

Pipeline	Group	N	Tests	Pearson q<.05	Spearman q<.05
GSR	ASD	69	87	26	16
GSR	Control	71	87	21	18
noGSR	ASD	69	87	12	14
noGSR	Control	71	87	0	6

BH is applied separately over 87 features per pipeline/diagnosis/estimator, following the original descriptive Pearson convention. These counts include algebraically related features and are not independent biological discoveries or group contrasts. All coefficients and alternative pooled corrections are in S2 Data.

H–efficiency relationships after nuisance adjustment

ASD partial H–global-efficiency correlations are .017 with GSR and .077 without GSR ($n = 66$); neither is significant after correction. Diagnosis-by-efficiency interactions are also nonsignificant (Holm $p = .635$ and .940), providing no support for an ASD-specific trade-off. Figure 7 displays the residualized global-efficiency relationships. Control partial correlations are .186 with GSR and $-.122$ without GSR ($n = 69$). The eight global/local efficiency tests yield no BH-corrected association. The GSR interaction coefficient is -8.58 (95% CI -25.49 to 8.34); the no-GSR coefficient is $-.65$ (-17.78 to 16.48). These intervals allow effects in either direction and do not support a diagnosis-specific relationship.

Fig 7. H and global efficiency after nuisance adjustment. Residuals adjust for mean Fisher FC, mean FD, age, IQ and site separately within diagnosis and pipeline. Within-group singleton sites are omitted, leaving 66 ASD and 69 controls. Panel probabilities use BH across all eight global/local efficiency tests. No adjusted association survives. A significant association in one group alone would not establish a difference between groups.

Correlation analysis of network metrics with behavioral scores

Table 7 and Fig. 8 summarize the clinical analyses. All five original ADI-R/ADOS outcomes are retained with score-specific missingness and a common 88-feature schema across pipelines. Pearson and Spearman

analyses are both reported. No clinical feature correlation survives FDR; adjusted ADOS-total correlations with H are .008 and .061 ($n = 50$).

Table 7. Adjusted H associations and clinical prediction.

Pipeline	Score	N	Partial r	BH q	OOF R2	MSE ratio
GSR	ADI R Social	30	0.078	0.961	-0.416	1.360
GSR	ADI R Verbal	30	0.203	0.961	-0.130	1.099
GSR	ADI R Onset	28	-0.026	0.961	-0.928	1.741
GSR	ADOS Total	50	0.008	0.961	-0.029	1.011
GSR	ADOS Communication	43	0.009	0.961	-0.047	1.012
noGSR	ADI R Social	30	0.354	0.528	-0.381	1.325
noGSR	ADI R Verbal	30	0.420	0.514	-0.370	1.332
noGSR	ADI R Onset	28	0.153	0.961	-0.348	1.217
noGSR	ADOS Total	50	0.061	0.961	-0.105	1.085
noGSR	ADOS Communication	43	0.059	0.961	-0.305	1.262

The partial correlation controls age, FIQ, mean FD and site; BH covers the ten score/pipeline tests. PLSR uses the common 88-feature set and nested site-held-out validation. MSE ratio compares PLSR with the corresponding outer-training-mean predictor; ratios above one favor that baseline.

Fig 8. Clinical associations and held-out score prediction. Left: adjusted H partial correlations with five clinical outcomes; no BH-corrected association is present. Right: nested site-held-out PLS regression R-squared for the common 88-feature schema. All R-squared values are negative. Both panels concern ASD participants; magenta therefore denotes ASD throughout, with filled/open symbols or hatching distinguishing GSR/no-GSR.

Multivariate analysis: Partial Least Squares. PLSR and PLSC remain distinct complements, as in the original paper. Score-specific PLSR samples range from 28 to 50; every OOF R^2 is negative and every MSE exceeds its training-mean baseline. No PLSC test survives BH correction; a nominal no-GSR five-score LV3 probability of .007 becomes $q = .098$ and is not presented as a confirmed latent association.

Machine learning classification of ASD versus controls

The 10 completed classifier/pipeline pairs have 140 unique OOF predictions each, using the same frozen site-disjoint splits. Completed algorithms are logistic regression, RBF SVM, Gaussian naive Bayes, KNN, XGBoost. Balanced accuracy ranges from 0.457 to 0.558 (Table 8, Fig. 9). Every conditional interval includes .5. No final classifier is selected from outer-fold performance. The conditional intervals do not include all training or QC-selection uncertainty.

Fig 9. Site-held-out ROC curves for all five classifier families. Curves use complete pooled out-of-fold scores for 140 participants per pipeline. Model colors are consistent across panels and follow the original classifier palette. The dashed diagonal indicates nondiscrimination. Each participant is scored by a model trained without that participant or acquisition site; the upstream node mask remains conditional on cohort-wide QC.

Figure 10 summarizes balanced accuracy and its conditional uncertainty.

Fig 10. Balanced accuracy and conditional uncertainty. Each classifier has complete out-of-fold predictions for all 140 participants. Symbols distinguish GSR and no-GSR; colors identify model families. Intervals are 95 percentiles from 1,000 site-bootstrap replicates conditional on the saved predictions and omit full retraining and upstream QC uncertainty. Every interval includes .5. No final model is selected from these outer results.

Table 8. Complete site-held-out classification results.

Pipeline	Model	Accuracy	BA	BA 95% CI	F1	AUC
GSR	GNB	0.500	0.498	0.428, 0.583	0.435	0.513
GSR	KNN	0.521	0.518	0.455, 0.581	0.385	0.505
GSR	LR	0.536	0.534	0.469, 0.613	0.488	0.535
GSR	SVM	0.557	0.558	0.464, 0.653	0.569	0.573
GSR	XGB	0.457	0.457	0.390, 0.531	0.441	0.485
noGSR	GNB	0.514	0.515	0.401, 0.633	0.528	0.499
noGSR	KNN	0.543	0.542	0.470, 0.627	0.522	0.518
noGSR	LR	0.486	0.485	0.428, 0.559	0.463	0.451
noGSR	SVM	0.486	0.485	0.385, 0.591	0.471	0.512
noGSR	XGB	0.507	0.507	0.439, 0.615	0.481	0.536

Every row uses 140 OOF predictions (69 ASD/71 controls). BA: balanced accuracy; F1 treats ASD as positive. Intervals bootstrap sites conditional on fixed predictions and omit part of training uncertainty. LR: logistic regression; SVM: radial-basis support vector machine; GNB: Gaussian naive Bayes; KNN: k-nearest neighbors; XGB: XGBoost. All models are reported without an outer-test winner.

Feature-selection audit and exploratory extension. Feature selection was already nested in the five-model analysis above. Reconstructing all 50 outer-training fits reproduces every saved prediction: ANOVA retained 22 features in 11 fits, 44 in 17, and all 88 in 22. We subsequently compared broader selection methods in a separate exploratory analysis, retaining the same known outer site folds rather than treating them as a fresh independent test. Training-fold near-exact redundancy pruning left 59 distinct columns, and inner CV chose all 59 in seven fits, ANOVA top-K in 16, mutual-information top-K in 17, and L1-ranked top-K in ten. Additional selection retained 5, 10 or 20 features.

Four of the ten model/pipeline point estimates increased and six decreased (Table 9). No-GSR logistic regression changed from .485 to .544, no-GSR SVM from .485 to .536, and no-GSR KNN from .542 to .557. GSR XGBoost changed from .457 to .492. Every paired conditional interval for change includes zero, and all follow-up balanced-accuracy intervals include .5. The follow-up therefore does not establish a reliable performance gain or justify selecting a new final classifier from the outer results. Full ROC, confusion, performance and selected-feature frequency heatmaps are in S1 Appendix; all feature names, inner candidates, and test predictions are in S2.

Table 9. Exploratory feature-selection comparison on the same known outer folds.

Pipeline	Model	Baseline BA	Follow-up BA	Change	Change 95% CI
GSR	GNB	0.498	0.491	-0.008	-0.092, 0.074
GSR	KNN	0.518	0.442	-0.076	-0.193, 0.024
GSR	LR	0.534	0.521	-0.013	-0.107, 0.036
GSR	SVM	0.558	0.534	-0.023	-0.097, 0.068
GSR	XGB	0.457	0.492	0.036	-0.031, 0.103
noGSR	GNB	0.515	0.493	-0.022	-0.086, 0.038
noGSR	KNN	0.542	0.557	0.014	-0.098, 0.108
noGSR	LR	0.485	0.544	0.059	-0.015, 0.109
noGSR	SVM	0.485	0.536	0.051	-0.038, 0.148
noGSR	XGB	0.507	0.500	-0.006	-0.149, 0.078

Paired change intervals resample sites conditional on fixed OOF predictions. The follow-up was specified after baseline results had been examined and is not independent confirmation of improvement. Representation and XGBoost search grids also differ. All ten estimates are shown; no classifier is selected by outer performance.

Sensitivity analysis: results without global signal regression

The original no-GSR sensitivity remains explicit at each scale: global quantities, within-network and between-network group contrasts, H–metric associations, clinical analyses, PLS, and classification. Primary participant and node sets are identical across pipelines. GSR/no-GSR Hamiltonian diagnosis estimates are both imprecise; ComBat and pooled trimming do not yield a corrected group difference. Larger-node sensitivity designs change participants as well as nodes. The original spectral-bipolarity descriptor is now reported separately as an exploratory raw adjusted comparison: coefficients are .00040 with GSR and .01055 without GSR, with Holm $p = .916$ for both. It remains outside the frozen feature families and does not alter the common clinical/predictive schema.

All five original algorithm types have complete OOF predictions in both pipelines, using the same revision participants, features, folds and scoring. Matching the algorithm types does not restore identical original inputs or validation: site-held-out evaluation replaces the original random train/test procedure.

Discussion

The revised analysis does not detect a corrected diagnosis effect in the coverage-defined adult male sample. Raw covariate-adjusted models, explicit ComBat sensitivities, pooled trimming, and mixed-diagnosis-site restriction are consistent in this limited sense. The separate male-control null comparison supports alignment relative to the specified state randomizations, while the diagnosis comparison asks a different question. Alignment in controls can coexist with no detectable diagnosis contrast. Their uncertainty remains substantial, and inconclusive equivalence tests preclude a universal adult-null interpretation. The

multisite generalization difficulties reported for FC more broadly [10] reinforce the need for larger independent tests.

This result concerns a relative fALFF–FC alignment statistic. Median states do not distinguish globally elevated from globally reduced fALFF distributions, and identical values can receive different labels across participants. It is therefore inappropriate to equate +1 with high neuronal firing or to interpret H as metabolic cost. Construction also links the statistic to global weights; the original descriptive H–metric question still produces corrected within-group associations, whereas the controlled efficiency analyses do not retain the submitted ASD-specific trade-off interpretation. An association between two summaries of the same FC matrix would not independently validate a biological mechanism even if statistically detected.

The result differs in scope from our earlier developmental triadic-balance work [8] and from temporal energy-landscape studies [9]. Different ages, state definitions, parcellations, and statistics can test different hypotheses. Regional adult fALFF findings [2] can coexist with a null global rank-based summary. We cannot infer developmental normalization or disagreement with dynamic-state findings from this adult cross-sectional analysis.

Regional organization and mathematical dependence

The lack of a corrected global diagnosis effect could, in principle, conceal opposing regional changes. The within- and between-network analyses explicitly address that possibility: all available cortical blocks are tested rather than choosing a few networks after inspecting their nominal probabilities. Their uncorrected contrasts do not survive the specified family corrections. This limits the present evidence for spatially localized group effects; it does not show that each cortical system is equivalent between groups. In particular, the single retained limbic parcel makes within-limbic comparisons unavailable, so a limbic null conclusion would be unsupported.

The descriptive H–metric analysis has a different outcome and purpose. Several within-group associations survive its stated correction, preserving an important part of the original study. Their interpretation depends on construction. Block energy is a component of the global sum, and multiplying mean state-weighted FC by a fixed negative edge count reproduces that block's energy. Consequently, significant correlations between these measures can describe how a global statistic is distributed across networks without constituting independent replications or identifying a mechanism. Nor does a significant association in ASD and a nonsignificant association in controls establish that the two associations differ. The adjusted interaction tests directly address that stronger question for efficiency and provide no corrected support for it.

The empirical-null results also depend on what is held fixed. Both null schemes preserve every FC weight, the observed number of nodal labels, and the graph's topology. They test reassignment of regional states, not whether the empirical FC network is unusually organized compared with rewired graphs. Constraining permutations by network and hemisphere yields less extreme alignment scores, consistent with some organization already being retained by the null. Remaining alignment can reflect finer spatial structure, biological organization, or shared measurement characteristics. A null model defines the comparison rather than furnishing a universal test of randomness [13].

Clinical relevance and predictive evaluation

Clinical correlation, latent covariance, and held-out prediction answer different questions. A PLSC component can summarize covariance within a complete-case sample without predicting an unseen person's score. PLS regression instead requires useful out-of-sample prediction, which was not observed here: all score-specific models have negative OOF R-squared and greater squared error than their corresponding training-mean predictors. Missing clinical scores further reduce the sample and can select a nonrepresentative subgroup. Neither the univariate results nor the latent-variable results justify a clinical severity marker in this cohort.

The classification analysis retains all five original algorithm families, including nonlinear models, and fits preprocessing and selection within training folds. The broad confidence intervals and weak ROC discrimination provide no evidence for a clinically useful diagnostic system. Some feature selection follow-up point estimates improve, but the paired intervals include no improvement. Because those methods were examined after the baseline outer results were known, selecting the best follow-up model would introduce another layer of selection. Independent evaluation is needed before attributing a reproducible gain to a selector or classifier. Feature-selection frequency likewise records which variables a procedure retained across overlapping training samples; it does not measure causal relevance, inferential significance, or stable biomarker importance.

Sex and generalizability

The male-only design restricts both the population represented and the questions that can be answered. The selected ABIDE pool does not support an inference about autistic women, girls, nonbinary people, or people with intellectual disability. The practical scarcity of eligible female participants in a dataset does not justify extending male estimates to all autistic people. Repeated male-only sampling can leave diagnostic frameworks and candidate biomarkers poorly tested in underrepresented groups and may

contribute to their under-recognition.

Sex-dependent diagnostic FC patterns have been reported [14]. Sex-related variation in either FC or fALFF could affect their joint alignment, even when within-person median splitting removes absolute fALFF scaling. Consequently, neither this group-null result nor the absence of a corrected efficiency association can be assumed to generalize to autistic females. Recorded binary sex metadata also do not capture gender identity or the social and clinical pathways influencing ascertainment. An inclusive test should recruit sufficient participants across sexes, ages, cognitive abilities, and presentation profiles; prespecify sex-by-diagnosis analyses and measurement-QC comparisons; and validate calibration and performance in each group. A small female subgroup added only for an underpowered post hoc comparison would not resolve this gap.

Other limitations and revision transparency

Coverage selection removes 94 CC200 nodes from the primary graph and leaves only one limbic node; the result is not a complete whole-brain or limbic test. The 50% support rule can still permit heterogeneous spatial sampling within a parcel. Larger-node designs trade coverage for participant retention and do not isolate a node-count effect. Signed associations depend on nuisance processing, and unmodeled site/acquisition effects may remain. HC3 addresses leverage and heteroscedasticity at participant level; it is not site-clustered uncertainty. Clinical samples are small, permutation assumptions are approximate, and coarse block nulls do not preserve full spatial autocorrelation [15]. Prediction is conditional on cohort-wide QC.

The original within-ASD no-GSR H–efficiency association can be reproduced from the supplied original feature exports ($r = -.324$, $q = .00592$, $n = 80$). Its absence in the revision's 106-node definition does not by itself establish that the original association was false: the participants, nodes, fALFF inputs and model assumptions differ. Nor does reproduction of that association establish a between-diagnosis interaction or independent biological mechanism. We preserve both the original record and the revised analyses, stating their different scopes.

The original-code audit identified discrepancies in cohort flow, ordinal cortical assignments, a fixed-TR spectral ratio calculated from filtered ROI series, and exports labeled ComBat that equaled raw values after a failed harmonization path. The revised analysis uses downloaded fALFF maps, explicit anatomical assignments, and checked harmonization. Multiple elements changed together, so we cannot attribute changed results solely to correction of site effects. The analysis uses the publicly distributed PCP fALFF maps as provided. We did not generate their voxelwise spectral numerator or denominator. Detailed

historical implementation choices are therefore inherited from the derivative provider rather than independently reconstructed here. This is a limitation of upstream provenance, not evidence that the downloaded maps were calculated incorrectly. Physiological interpretation remains restricted to a relative alignment statistic derived from those maps.

Conclusions. Within the specified coverage-limited network of adult males, the revised analyses provide no corrected evidence of diagnosis differences or clinical/predictive utility. Male controls show alignment relative to the specified state-permutation nulls, a baseline property distinct from diagnosis discrimination. These analyses do not establish group equivalence, a universal autism null, or an ASD-specific energy–efficiency mechanism. Independent, inclusive validation and more complete characterization of upstream derivatives would strengthen assessment of the framework.

Methods

Data source, eligibility, and ethics

The source pool comprises 182 previously selected ABIDE I participants (90 ASD, 92 controls) with requested PCP derivatives [16, 17]. We did not reconstruct an initial random or matched 200-person sample. Eligibility for the revision required age 18–30 inclusive and observed FIQ > 80, without IQ imputation, yielding 140 participants (69 ASD, 71 controls) at 12 sites. All were recorded male. We did not impose a right-handedness restriction: 71 were recorded right-handed, six left-handed, three ambidextrous, and 60 missing or sentinel-coded. Existing diagnoses were taken from ABIDE metadata; we do not retrospectively assign DSM-5 ascertainment to all contributing sites. Clinical missingness was handled per outcome rather than excluding everyone missing any clinical score.

The sample flow is shown in Fig. 1. Model-specific counts and covariates are given in Table 1; site counts and participant ledgers are supplied separately in S1 Appendix and S2 Data.

The original ABIDE studies were approved by the contributing sites' institutional review boards, with written informed consent and/or assent obtained under their respective protocols [16]. The present study exclusively reanalyzed publicly available, deidentified ABIDE data. No new participants were recruited, no identifying information was accessed, and there was no participant contact or new data collection. No additional ethics approval or consent was obtained for this secondary analysis because it used only those public, deidentified records.

Downloaded derivatives and upstream processing

Regional CC200 time series, fALFF maps, and functional masks were downloaded from the PCP C-PAC outputs. MRI preprocessing, time-series extraction, and voxelwise fALFF computation were performed upstream, not locally for this revision. The PCP documentation describes slice-time correction, motion correction, intensity normalization, and nuisance regression using six motion parameters, their one-time-point lags, the squares of those 12 terms, five white-matter/CSF principal components, and linear and quadratic trends. The global signal is included only in the GSR strategies [17]. We use filtered ROI time series from `filt_global` and `filt_noglobal` for FC, and maps and masks from the corresponding `nofilt` strategy for fALFF aggregation. The documented FC filtering band is 0.01–0.1 Hz; no additional local temporal filter or fixed-length truncation was applied. Primary time-series lengths range from 116 to 296 volumes.

TR was recovered from inherited site acquisition metadata, including a FILE_ID-specific site-variant resolution where necessary. Among the 140 eligible participants, TR is 1.5 s ($n = 19$), 1.667 s ($n = 16$), 2.0 s ($n = 87$), 2.2 s ($n = 1$), or 3.0 s ($n = 17$). Consequently, a universal 0.25-Hz Nyquist limit is inappropriate. Mean FD comes from PCP summary metadata; per-volume censoring histories are not reconstructed from those summaries.

Provider-derived fALFF and scope of local processing. We used the voxelwise fALFF maps distributed by PCP for ABIDE I, generated upstream with C-PAC. Our local processing extracted regional means from these maps; it did not compute voxelwise fALFF, its spectral denominator, or a new amplitude ratio from the filtered FC time series. The provider's archived configuration lists 0.01–0.1 Hz fALFF bounds [18], and its output mapping identifies the distributed map as a smoothed standard-space fALFF derivative [19]. Accordingly, the former description of a locally applied Butterworth filter followed by a 0.01–0.08 Hz numerator over a fixed 0–0.25 Hz denominator does not describe the present analysis and is not retained. The downloaded maps are used without local spectral recalculation, including for participants with TR of 3 s.

The precise historical denominator implementation, input filtering order, and smoothing kernel were not independently established from the available release records. We therefore attribute the upstream computation to PCP/C-PAC and do not claim to have verified an unfiltered denominator. Selecting a `nofilt` strategy directory alone does not establish that implementation. This reporting limitation does not itself demonstrate an error in the provider's maps, but it limits claims about their exact spectral construction. The regional aggregation, coverage rules, state definition, FC calculation, and subsequent statistics are specified below and supplied in S3 Code.

Coverage-defined nodes and cortical assignments

CC200 parcel labels were resampled by nearest-neighbor interpolation to each fALFF map grid. Regional fALFF is the arithmetic mean of finite map values within the parcel and the downloaded functional mask; coverage is the fraction of parcel voxels meeting these criteria. A retained node must have a nonconstant downloaded time series and at least 50% finite in-mask support in both pipelines. The primary set contains the 106 nodes satisfying this rule in every eligible participant. All 140 participants therefore have the same primary nodes. This is a coverage-limited network rather than an unchanged whole-brain CC200 graph. The threshold is a methodological choice, not a universal validity standard.

Two exploratory designs assess scope: 181 nodes qualifying in at least 95% of eligible participants, followed by requiring every selected node to qualify in each retained participant (104 participants, 48 ASD/56 controls); and all 200 nodes with the same participant-level rule (69 participants, 31/38). Medians and all features are recomputed within each node set. These designs change both nodes and participants, precluding attribution of between-design differences to anatomy alone.

Cortical network labels use the liberal Yeo seven-network atlas [20]. Parcels require at least 50% Yeo cortical support and are then assigned by maximum Dice overlap. Other nodes are unassigned for cortical blocks but retained globally. The 106-node set includes 91 assigned and 15 unassigned nodes. Only one assigned node is limbic; within-limbic summaries are structurally unavailable, not null tests. The atlas mirror's inconsistent float32 bit-depth header was normalized after payload and label validation; voxel values and spatial transforms were preserved.

Hamiltonian and complementary features

For participant k, let a_{ik} be the regional downloaded fALFF mean and m_k the median over the selected nodes. We define

$$s_{ik} = \begin{cases} +1, & a_{ik} \geq m_k, \\ -1, & a_{ik} < m_k. \end{cases} \tag{1}$$

These are relative mathematical labels: the same fALFF value can receive opposite labels in two participants. An above-median label does not imply a higher mean neuronal firing rate. The median split also removes absolute between-person fALFF scaling from the state definition.

Pearson correlations of the downloaded filtered regional series are clipped to $[-0.9999, 0.9999]$, Fisher

transformed, and assigned zero diagonal. With these weights $z_{ij,k}$, the primary statistic is

$$H_k = -\sum_{i<j} z_{ij,k} s_{ik} s_{jk}. \tag{2}$$

In Eq. 2, lower values indicate greater alignment under this definition, not lower biological energy or demonstrated dynamical stability. The four pairwise sign/state classes are illustrated in Fig. 11. Negative FC is a statistical association, not a direct inhibitory synapse. We retain GSR and no-GSR as paired analyses because GSR affects signed associations and can alter group comparisons [21]; GSR is not treated as biologically definitive.

Fig 11. Pairwise state and connection configurations. Nodes use the original magenta and dark-green state palette for relative labels +1 and -1, not diagnosis in this conceptual diagram. Positive and negative FC links combine with state agreement or opposition. Same-state positive and opposite-state negative connections lower H; the other two configurations raise it. Relative fALFF labels do not denote neuronal firing or excitation and inhibition.

Complementary features include four proportions partitioning nonzero edges by weight sign and nodal-state agreement, and global/local efficiency of the absolute Fisher-weight graph [22]. Edge lengths are $1/|z_{ij}|$, with zero weights treated as absent; local efficiency averages the induced-neighborhood global efficiencies. For each of seven within-network and 21 between-network blocks we compute mean weight, mean state-weighted weight, and their negative sum. Edges involving unassigned nodes remain in H but not cortical block sums. The previous second-eigenvector "bipolarity" descriptor is exported only as a legacy descriptor and is excluded from inference.

Group inference and harmonization

Primary inference uses untrimmed raw H in ordinary least squares with diagnosis, site indicators, age, FIQ, and mean FD. We use HC3 standard errors and t-referenced 95% confidence intervals. Each design excludes sites with fewer than three retained participants to avoid singleton leverage; the primary model includes 136 participants (67/69) at nine sites. Design rank is checked. Holm correction covers the two pipeline diagnosis tests. Secondary whole-network, within-network, and between-network families use Benjamini–Hochberg correction across both pipelines within each design/model, excluding unavailable tests and primary H. There are respectively 12, 36, and 126 valid secondary tests in the primary design. Sensitivity analyses are exploratory rather than further confirmatory primary families.

ComBat adjusts the valid feature matrix with site as batch, explicit categorical diagnosis protection, and continuous age, FIQ, and FD protection [23, 24]. The same adjusted regression is then fitted. Failures cannot silently substitute raw values. Uncertainty does not integrate estimated harmonization parameters.

Pooled 1.5-IQR trimming of raw H, before reapplying the site-size rule, is a separate sensitivity; no diagnosis-specific trimming defines the primary analysis or prediction sample.

Additional reviewer diagnostics summarize site/group H, leverage and Cook's distance, and site partial R^2 conditional on biological covariates. Residual variance comparisons and post-ComBat site summaries are descriptive because the same sample estimates harmonization. An exploratory raw adjusted analysis excludes sites with only one diagnosis, with Holm correction over pipelines per design. No participant is removed on the basis of influence diagnostics.

We report unadjusted Hedges' g with 2,000 within-group participant bootstrap replicates (seed 101). These intervals are descriptive and do not resample sites. Sensitivity power uses a two-sample approximation at 80% power, not a site/covariate-adjusted power calculation. Exploratory two-one-sided equivalence tests use bounds of ± 0.2 observed pooled SD for the adjusted coefficient; these are conventional bounds, not a validated clinically negligible effect.

Efficiency associations and randomization checks

Within-group partial correlations adjust for mean Fisher FC, FD, age, IQ, and site. Sites becoming singletons within a group are excluded from that group's residual analysis. Eight tests (two efficiency summaries, two groups, two pipelines) receive BH correction. Group specificity is tested directly by diagnosis-by-standardized- global-efficiency interaction with the same controls, Holm corrected over pipelines. A uniform edge shift c changes H by

$$\Delta H = \frac{n - (\sum_i s_i)^2}{2} c. \tag{3}$$

For exactly balanced 106-node states this is $53c$, demonstrating a mathematical route for global-weight shifts to affect H independently of a diagnostic mechanism.

For each participant and pipeline, 1,000 state permutations hold FC and total state counts fixed. Unrestricted permutations are compared with permutations within network/hemisphere blocks (hemisphere defined by parcel-centroid $x < 0$ versus $x \geq 0$). We compute participant z-scores and lower-tail Monte Carlo probabilities with a plus-one correction; zero null variance is unavailable. Seeds are $10000 + \text{SUB_ID} + 1$ for GSR and $10000 + \text{SUB_ID}$ for no-GSR. Block permutations preserve coarse organization, not full spatial autocorrelation [15]; they are alignment checks rather than independent validation or diagnosis tests.

To report a non-autistic baseline separately, we summarize these existing null outputs for all 71 eligible

male controls, without energy trimming or the regression-specific small-site exclusion. Individual lower-tail probabilities receive exploratory BH correction over all 560 participant/pipeline/null comparisons, including both diagnostic groups. Mean energy differences and mean null z-scores have descriptive 95% percentile intervals from 5,000 site-cluster bootstrap replicates (seed 20261026), conditional on the saved null summaries. These intervals are not pooled randomization tests and omit Monte Carlo estimation uncertainty. The signed weighted FC matrix is fixed in these nulls; no edge/weight rewiring or independent edge-sign reassignment is performed.

Descriptive H–metric complement and original spectral descriptor

The original descriptive H-versus-feature question is retained as an additional revision complement. Pearson and Spearman correlations cover all 87 available nonlegacy features other than H, within each diagnosis and pipeline. BH is computed separately over 87 tests per group/pipeline/estimator, matching the original Pearson family convention. An alternative pooled BH over 348 tests per estimator/version is supplied separately; pooling families is not uniformly more conservative in every subgroup. Raw eligible samples, raw adjusted-model subsets, and saved ComBat features on the same model subsets are reported as distinct versions. These analyses do not change frozen group-inference families. The spectral bipolarity value from the original second-eigenvector construction is additionally compared by raw HC3-adjusted diagnosis models with Holm over two pipeline tests, separately from frozen families. It is an operational spectral descriptor, not an asserted optimal signed-balance partition.

Clinical and predictive analyses

The primary design supplies the same 88 available, nonlegacy features to both pipelines and all clinical/predictive feature analyses, including both efficiencies. Five ASD outcomes are considered: ADI-R social, verbal, and onset scores, ADOS total, and ADOS communication. Missing/sentinel scores are excluded per outcome. Pearson and Spearman feature correlations receive separate BH corrections across all feature/score/pipeline tests. Adjusted H correlations control age, IQ, FD, and site, with BH across available score/pipeline tests.

PLS regression uses nested site-disjoint GroupKFold (up to five outer and three inner folds), selecting one to three components by inner-fold MSE. Imputation, variance filtering, and PLS scaling are fitted inside training folds. Complete OOF predictions are compared with each outer training sample's mean outcome. Exploratory PLSC uses nuisance-residualized and standardized features/outcomes for ADOS and five-score complete-case blocks, with 1,000 within-site residual permutations (seed 101), rank-wise

singular-value tests, and BH over all reported latent-variable tests. Its residual exchangeability assumptions and small complete cases limit interpretation [25].

Classification compares logistic regression, RBF SVM, Gaussian naive Bayes, KNN, XGBoost on all 140 primary participants without ComBat. Deterministic size-balanced site-disjoint folds use five outer and three inner splits, each required to contain both classes. Median imputation, variance filtering, standardization, ANOVA feature selection (25/50/100 percent), and hyperparameter tuning are nested. Logistic and SVM class weights are balanced. Logistic C values are 0.01/0.1/1/10; SVM C values are 0.1/1/10 with scale-based gamma. Gaussian naive Bayes uses its default variance smoothing and training-estimated priors, with feature-selection percentile tuned inside folds. KNN evaluates 3/5/7/9/11 neighbors and uniform/distance weights. Positive-class probabilities supply ROC scores for these added models; logistic/SVM decision-function scores and predictions are reused unchanged. XGBoost uses CPU histogram trees, seed 42, one thread, log-loss evaluation, max depth 2/3/4, 100/200/300 estimators, and learning rates .05/.1. The same nested feature-selection percentiles apply; default scale-pos-weight is retained. Inner scoring uses balanced accuracy. No model is selected by outer-test accuracy. Pooled OOF balanced accuracy is accompanied by a 1,000-replicate site bootstrap conditional on fixed predictions, using the same seed 303 and site order as the completed Stage 2 models. This omits some training uncertainty. The node mask and available feature schema use cohort-wide QC, so prediction is exploratory conditional on this mask, not prospective validation of the entire workflow.

Exploratory comparison of feature-selection methods. After inspecting the baseline outer results, we specified a separate follow-up using the same 88-feature inputs, participants, and five/three outer/inner site folds. Imputation, variance filtering, redundancy pruning, scaling, selection, and classifier fitting are estimated only from the corresponding training data [26]. Redundancy pruning retains the first feature in fixed schema order when absolute training correlation is at least $1 - 10^{-10}$; proportional EnergyBlock/AvgFalffW terms and complementary proportions motivate this step. The graph, participants and primary group tests are unchanged.

Inner CV jointly chooses all remaining features, or ANOVA, mutual information, or L1-logistic coefficient-ranked top 5/10/20 features, with K capped at the available count. Mutual information uses three neighbors and seed 101. The L1 ranker uses balanced liblinear logistic regression, C=1 and seed 101; coefficient ties are broken by training ANOVA rank and fixed feature order. Zero-coefficient selections are explicitly recorded; none occurred in the final outer fits. Non-XGBoost classifier grids match the baseline. The compact follow-up XGBoost grid uses depth 2/3, estimators 100/200 and learning rates .05/.1, with

the same fixed CPU histogram settings. Thus changes cannot be attributed to selector type alone. All candidates are selected by inner balanced accuracy, and all ten model/pipeline results are reported without choosing an outer-test winner.

Paired 1,000-replicate site bootstrap (seed 303) estimates conditional intervals for the balanced-accuracy change on aligned OOF predictions. These intervals omit full training uncertainty and do not adjust for repeated exploration. Because baseline outer results had already been inspected, this follow-up is exploratory rather than independent confirmation of improved generalization. External evaluation would be needed to substantiate a predictive gain.

Data and code availability

ABIDE I participant data and PCP derivatives are available through the ABIDE and Preprocessed Connectomes Project repositories (`https://fcon_1000.projects.nitrc.org/indi/abide/` and `https://preprocessed-connectomes-project.org/abide/`). S2 Data supplies the revision feature tables, participant covariates used in the reported models, complete test results, out-of-fold predictions, selection records, and provenance manifests. S3 Code supplies the executable revision scripts, analysis specifications, and instructions for rebuilding the analyses from the source derivatives. S1 Appendix supplies additional figures and diagnostics. The source repositories govern access to the original participant imaging; those images are not redistributed in the supporting archives.

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References

1. Zou QHea. An improved approach to detection of amplitude of low-frequency fluctuation (ALFF) for resting-state fMRI: Fractional ALFF. Journal of Neuroscience Methods. 2008;172(1):137–141. doi:10.1016/j.jneumeth.2008.04.012.

2. Itahashi T, Yamada T, Watanabe H, Nakamura M, Ohta H, Kanai C, et al. Alterations of local spontaneous brain activity and connectivity in adults with high-functioning autism spectrum disorder. Molecular Autism. 2015;6:30. doi:10.1186/s13229-015-0026-z.

3. Rubinov M, Sporns O. Weight-conserving characterization of complex functional brain networks. NeuroImage. 2011;56(4):2068-79. doi:10.1016/j.neuroimage.2011.03.069.

4. Heider F. Attitudes and Cognitive Organization. The Journal of Psychology: Interdisciplinary and Applied. 1946;21(1):107–112. doi:10.1080/00223980.1946.9917275.

5. Cartwright D, Harary F. Structural balance: A generalization of Heider's theory. Psychological Review. 1956;63:277-93.

6. Kargaran A, Jafari GR. Heider and coevolutionary balance: From discrete to continuous phase transition. Physical Review E. 2021;103:052302. doi:https://doi.org/10.1103/PhysRevE.103.052302.

7. Ghanbarzadeh Noudehi M, Kargaran A, Azimi-Tafreshi N, Jafari GR. Second- to first-order phase transition: Coevolutionary versus structural balance. Phys Rev E. 2022;106:044303. doi:10.1103/PhysRevE.106.044303.

8. Moradimanesh Z, Khosrowabadi R, Gordji ME, Saberi M, Jafari G. Altered structural balance of resting-state networks in autism. Scientific Reports. 2021;11:1966. Available from: https://www.nature.com/articles/s41598-020-80330-0. doi:10.1038/s41598-020-80330-0.

9. Watanabe T, Rees G. Brain network dynamics in high-functioning individuals with autism. Nature Communications. 2017;8:16048. doi:10.1038/ncomms16048.

10. King JB, Prigge MBD, King CK, Morgan J, Weathersby F, Fox JC, et al. Generalizability and reproducibility of functional connectivity in autism. Molecular Autism. 2019;10:27. doi:10.1186/s13229-019-0273-5.

11. Nygaard V, Rødland EA, Hovig E. Methods that remove batch effects while retaining group differences may lead to exaggerated confidence in downstream analyses. Biostatistics. 2016;17(1):29-39. doi:10.1093/biostatistics/kxv027.

12. Holiga S, Hipp JF, Chatham CH, Garces P, Spooren W, Liogier D'Ardhuy X, et al. Patients with autism spectrum disorders display reproducible functional connectivity alterations. Science Translational Medicine. 2019;11(481):eaat9223. doi:10.1126/scitranslmed.aat9223.

13. Vasa F, Misic B. Null models in network neuroscience. Nature Reviews Neuroscience. 2022;23:493-504. doi:10.1038/s41583-022-00601-9.

14. Alaerts K, Swinnen SP, Wenderoth N. Sex differences in autism: a resting-state fMRI investigation of functional brain connectivity in males and females. Social Cognitive and Affective Neuroscience. 2016;11(6):1002-16. doi:10.1093/scan/nsw027.

15. Markello RD, Mišić B. Comparing spatial null models for brain maps. NeuroImage. 2021;236:118052. doi:10.1016/j.neuroimage.2021.118052.

16. Di Martino A, Yan CG, Li Q, Denio E, Castellanos FX, Alaerts K, et al. The Autism Brain Imaging Data Exchange: Towards a Large-Scale Evaluation of the Intrinsic Brain Architecture in Autism. Molecular Psychiatry. 2014;19(6):659-67. doi:10.1038/mp.2013.78.

17. Preprocessed Connectomes Project. ABIDE Preprocessed: C-PAC pipeline documentation [Internet]. Preprocessed Connectomes Project; n.d. [cited October 26, 2026]. Available from: https://preprocessed-connectomes-project.org/abide/cpac.html.

18. Preprocessed Connectomes Project. ABIDE rerun pipeline configuration and output mapping [Internet]. Preprocessed Connectomes Project; n.d. [cited October 26, 2026]. Available from: https://github.com/preprocessed-connectomes-project/abide/blob/master/preprocessing/settings/pipeline_config_abide_rerun.yml.

19. Preprocessed Connectomes Project. ABIDE derivative output mapping: map_rawout_to_organized_s3.py;. Accessed October 27, 2026. https://github.com/preprocessed-connectomes-project/abide/blob/master/map_rawout_to_organized_s3.py.

20. Yeo BTT, Krienen FM, Sepulcre J, Sabuncu MR, Lashkari D, Hollinshead M, et al. The organization of the human cerebral cortex estimated by intrinsic functional connectivity. Journal of Neurophysiology. 2011;106(3):1125-65. Available from: https://doi.org/10.1152/jn.00338.2011. doi:10.1152/jn.00338.2011.

21. Gotts SJ, Saad ZS, Jo HJ, Wallace GL, Cox RW, Martin A. The perils of global signal regression for group comparisons: a case study of Autism Spectrum Disorders. Frontiers in Human Neuroscience. 2013;7:356. doi:10.3389/fnhum.2013.00356.

22. Latora V, Marchiori M. Efficient Behavior of Small-World Networks. Physical Review Letters. 2001;87(19):198701. doi:10.1103/PhysRevLett.87.198701.

23. Johnson WE, Li C, Rabinovic A. Adjusting batch effects in microarray expression data using empirical Bayes methods. Biostatistics. 2007;8(1):118-27. doi:10.1093/biostatistics/kxj037.

24. Fortin JP, Cullen N, Sheline YI, Taylor WD, Aselcioglu I, Cook PA, et al. Harmonization of cortical thickness measurements across scanners and sites. NeuroImage. 2018;167:104-20. doi:10.1016/j.neuroimage.2017.11.024.

25. Krishnan A, Williams LJ, McIntosh AR, Abdi H. Partial Least Squares (PLS) methods for neuroimaging: A tutorial and review. NeuroImage. 2011;56(2):455-75. doi:10.1016/j.neuroimage.2010.07.034.

26. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research. 2011;12:2825-30. Available from: `http://www.jmlr.org/papers/volume12/pedregosa11a/pedregosa11a.pdf`.

Supporting information captions

S1 Appendix. Supporting methods, site/cohort diagnostics, node-set sensitivities, clinical latent-variable results, complete classifier confusion matrices, feature-selection frequencies, and descriptive PCA. Separate PDF.

S2 Data. Machine-readable features, analysis covariates and clinical outcomes, complete statistical results and predictions, input provenance and file-integrity manifests. ZIP archive with an indexed README.

S3 Code. Revision analysis and figure-generation source, frozen analysis specifications, execution instructions and software-version records. ZIP archive. Historical source notebooks are retained separately as audit material.